

Smart Health & Fitness API

API Endpoints Summary — COM661 Coursework 1

Base URL: `http://127.0.0.1:5001/api` | Authentication: Bearer JWT Token | Content-Type: application/json

1. Authentication

Method	Endpoint	Auth?	Description
POST	/register	No	Register a new user account
POST	/login	No	Authenticate and receive JWT access token

2. User Profile

Method	Endpoint	Auth?	Description
GET	/profile	Yes	Retrieve the authenticated user's profile
PUT	/profile	Yes	Update profile fields (name, age, weight, etc.)
DELETE	/profile	Yes	Permanently delete user account

3. Activity Logs

Method	Endpoint	Auth?	Description
POST	/activities	Yes	Log a new workout activity
GET	/activities	Yes	List activity logs (filter by type/date, paginated)
GET	/activities/{id}	Yes	Retrieve a single activity log entry
PUT	/activities/{id}	Yes	Update an existing activity log entry
DELETE	/activities/{id}	Yes	Delete an activity log entry

4. Nutrition Logs

Method	Endpoint	Auth?	Description
POST	/nutrition	Yes	Log a new meal / nutrition entry
GET	/nutrition	Yes	List nutrition logs (filter by meal type/date, paginated)
GET	/nutrition/{id}	Yes	Retrieve a single nutrition log entry
PUT	/nutrition/{id}	Yes	Update an existing nutrition log entry
DELETE	/nutrition/{id}	Yes	Delete a nutrition log entry

5. Health Metrics

Method	Endpoint	Auth?	Description
GET	/health-metrics	Yes	Get current health metrics (includes computed BMI)
PUT	/health-metrics	Yes	Update health metrics (heart rate, BP, sleep, VO2 max)

6. Fitness Goals

Method	Endpoint	Auth?	Description
GET	/goals	Yes	Retrieve the user's fitness goals
PUT	/goals	Yes	Set or update fitness goals
DELETE	/goals	Yes	Clear all fitness goals

7. Analytics

Method	Endpoint	Auth?	Description
GET	/analytics/bmi-report	Yes	BMI calculation with category and advice
GET	/analytics/calorie-balance	Yes	Calorie intake vs burned for date range
GET	/analytics/weekly-progress	Yes	Day-by-day breakdown for last 7 days
GET	/analytics/goal-tracking	Yes	Progress vs fitness goals
GET	/analytics/activity-summary	Yes	Aggregation pipeline: stats by activity type
GET	/analytics/nutrition-summary	Yes	Aggregation pipeline: macros by meal type
GET	/analytics/admin/top-active-users	Admin	Rank users by calories burned (Admin only)

8. Admin — User Management

Method	Endpoint	Auth?	Description
GET	/users	Admin	List all users with pagination and filters
GET	/users/{id}	Admin	Get a specific user by ID
DELETE	/users/{id}	Admin	Delete any user account
PUT	/users/{id}/role	Admin	Change a user's role (user / admin)

9. HTTP Status Codes Used

Code	Status	When Used
200	OK	Successful GET, PUT, DELETE operations
201	Created	Successful POST — new resource created
400	Bad Request	Missing/invalid fields, validation errors
401	Unauthorised	No token or invalid credentials
403	Forbidden	Authenticated but insufficient role (admin required)
404	Not Found	Resource does not exist
409	Conflict	Duplicate email on registration

10. Query Parameters

Endpoint	Parameter	Type	Description
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/activities (GET)	activity_type	string	Filter by activity type (e.g. running, cycling)
/activities (GET)	start_date / end_date	YYYY-MM-DD	Filter by date range
/activities (GET)	page / per_page	integer	Pagination (default: page=1, per_page=10)
/nutrition (GET)	meal_type	string	Filter by meal type (breakfast/lunch/dinner/snack)
/nutrition (GET)	start_date / end_date	YYYY-MM-DD	Filter by date range
/nutrition (GET)	page / per_page	integer	Pagination
/analytics/calorie-balance	start_date / end_date	YYYY-MM-DD	Date range (default: last 7 days)
/users (GET)	page / per_page	integer	Pagination (Admin)
/users (GET)	membership_type	string	Filter by membership (free/premium) (Admin)
/users (GET)	search	string	Search by full_name (Admin)
/analytics/admin/top-active-users	limit	integer	Max users returned (default: 10) (Admin)